

CBCS SCHEME - Make-Up Exam

[illegible]

BCS302

Third Semester B.E/B.Tech. Degree Examination, June/July 2025
Digital Design and Computer Organization

Time: 3 hrs.

Max. Marks:100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
1	a.	Apply the basic theorems and postulates of Boolean algebra to simplify the following expression to minimum number of literals and draw the original and simplified expressions with logic diagram. i) $xy + x^1z + yz$ ii) $x + x^1y$	10	L3	CO1
	b.	Apply k – map Technique to simplify the function $F (w, x, y, z) = \sum (1, 3, 7, 11, 15)$ and $d (w, x, y, z) = \sum (0, 2, 5)$ and also implement the simplified function with NAND gate.	10	L3	CO1
OR					
2	a.	Describe Map Method for three variables.	7	L2	CO1
	b.	Explain Positive and Negative logic and prove that positive AND is equal to negative OR.	7	L2	CO1
	c.	Develop a verilog gate –level description of the circuit shown in Fig. Q. 2 (c)	6	L3	CO1
Fig. Q. 2(c)					
Module – 2					
3	a.	Design a combinational circuit that converts a four – bit BCD code to Excess 3 code.	10	L2	CO3
	b.	Design 3 to 8 decoder. Also implement full adder using 3 to 8 decoder.	10	L3	CO2
OR					
4	a.	Describe 4×1 mux with block diagram and truth table. Also develop a behavioral model verilog code for 4×1 mux.	10	L3	CO2
	b.	What is flip flop? Explain positive edge triggered D type of flip flop.	10	L2	CO2
Module – 3					
5	a.	With a neat block diagram explain the functional units of computer.	10	L2	CO3
	b.	Explain the Basic Instruction types with examples.	10	L2	CO3

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OR					
6	a.	Define addressing mode. Explain any four types of addressing modes with example.	10	L2	CO3
	b.	Develop a program to add list of N numbers using auto increment addressing mode and develop the program for the concept of branching with relevant memory diagram for both data and program.	10	L3	CO3
Module – 4					
7	a.	With a neat diagram explain the concept of accessing I/O devices.	10	L2	CO4
	b.	What is bus arbitration? Explain centralized and distributed arbitration method with a neat diagram.	10	L2	CO4
OR					
8	a.	With neat sketches, explain various method for handling multiple interrupt requests raised by multiple devices.	10	L2	CO4
	b.	What is cache memory? Explain any two mapping functions of cache memory.	10	L2	CO4
Module – 5					
9	a.	Draw the single bus architecture and write the control sequence for execution of instruction ADD (R ₃) , R ₁	10	L3	CO5
	b.	With a suitable diagram explain the concept of register transfer and fetching of word from memory.	10	L2	CO5
OR					
10	a.	With a neat diagram explain the flow of 4 – stage pipeline operation.	10	L2	CO5
	b.	Explain the role of cache memory and pipeline performance.	10	L2	CO5
